

Capstone Project Hotel

Booking Analysis - EDA

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EDA (Exploratory Data Analysis)

Data Visualization, Overall Stats and **Conclusion**

Agenda of Data Analysis

To discuss the analysis of given hotel bookings data set from 2015-2017.

We'll be doing analysis of given data set in following ways :

- ❖ **Univariate Analysis**
- ❖ **Bivariate Analysis**
- ❖ **Hotel-wise Analysis**
- ❖ **Room Type Analysis**
- ❖ **Distribution Channel-wise Analysis**
- ❖ **Meal Type Analysis**
- ❖ **Booking Cancellation Analysis**
- ❖ **Time-wise Analysis (Monthly, Weekly, Yearly)**
- ❖ **Customer Type Analysis (Repeat vs. New Guests)**
- ❖ **Car Parking Demand Analysis**
- ❖ **Stay Length Analysis**
- ❖ **ADR Trends Analysis (Seasonality)**
- ❖ **Correlation Analysis**

By doing this we'll try to find out key factors driving the hotel bookings trends.

Hotel Booking Data

Introduction

This Dataset contains data that compares various booking information between two hotels, City Hotel and Resort Hotel. So, here I will be using the data to analyze the factors affecting the hotel bookings. These factors can be used for reporting trends and predict the future bookings.

This Dataset contains the booking data of hotels from year 2015-2017.

Data Description

hotel	There are two types of hotels, one is City Hotel and another is Resort Hotel
is_canceled	Here 0 and 1 value indicates booking was cancelled (1) or not (0)
lead_time	Time-lapse between reservation and actual arrival date
arrival_date_year	Year of arrival date
arrival_date_month	Month of arrival date
arrival_date_week_number	Week number of arrival date
arrival_date_day_of_month	Day of arrival date
stays_in_weekend_nights	Number of weekend nights (Saturday or Sunday) spent at the hotel by the guests

Data Description

stays_in_week_nights	Number of weeknights (Monday to Friday) spent at the hotel by the guests
adults	Number of adults among guests
children	Number of children among guests
babies	Number of babies among guests
meal	Type of meal booked
country	Country of guests
market_segment	Designation of market segment
distribution_channel	Name of booking distribution channel

Data Description

is_repeated_guest	If the booking was from a repeated guest (1) or not (0)
previous_cancellations	Number of previous bookings that were cancelled by the customer prior to the current booking
previous_bookings_not_canceled	Number of previous bookings not cancelled by the customer prior to the current booking
reserved_room_type	Code of room type reserved
assigned_room_type	Code of room type assigned
booking_changes	Number of changes/ amendments made to the booking
deposit_type	Type of the deposit made by the guest
agent	ID of travel agent who made the booking

Data Description

company	ID of the company that made the booking
days_in_waiting_list	Number of days the booking was in the waiting list
customer_type	Type of customer, assuming one of four categories
adr	Average Daily Rate, as defined by dividing the sum of all lodging transactions by the total number of staying nights
required_car_parking_spaces	Number of car parking spaces required by the customer
total_of_special_requests	Number of special requests made by the customer
reservation_status	Reservation status (Canceled, Check-Out or No-Show)
reservation_status_date	Date at which the last reservation status was updated

Data Summary

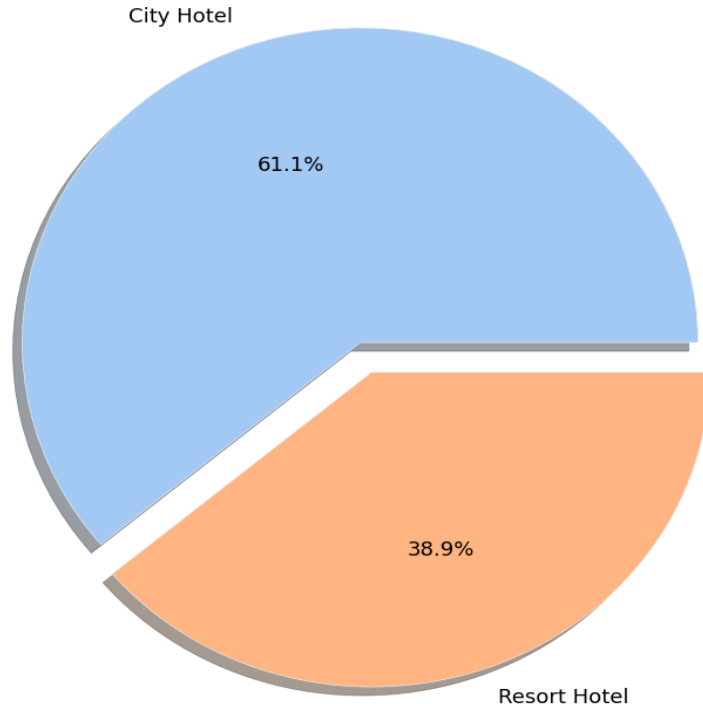
- This **data set** contains a single file which compares various **booking information** between **two hotels: City Hotel** and **Resort Hotel**. Includes information such as when the booking was made, **length of stay**, the number of **adults**, **children**, and/or **babies**, and the number of available **parking spaces**, among other things.
- The dataset contains a total of **119390** rows and **32** columns.
- All the columns are divided into three dtypes : **Object**, **float64** and **int64**.
- This dataset does have **duplicated values** as well as **null values**. There are total of **31994 duplicate values** and **four columns** have **missing values/ null values**.
- The **maximum** number of **missing values** are from '**Company**' column then followed by '**Agent**', '**Country**' and '**Children**' columns. The '**Children**' column consists of only **4 null values**, while '**Company**' column consists of **112593 null values**.

Points for Discussion

- Top Most Preferred Hotel & Hotel Rooms
- ADR (Average Daily Rate)
- Repeated Vs Non-Repeated Guests
- Requirement of Car Parking Space
- Most Preferred Meal
- Distribution Channel Vs ADR
- Top Booking Months & Year
- Optimal Stay Length
- Confirmation Vs Cancellation
- Mostly Arrived Customers/ Visitors
- Overall Stats
- Conclusion

Top Most Preferred Hotel

Most preferred Hotels

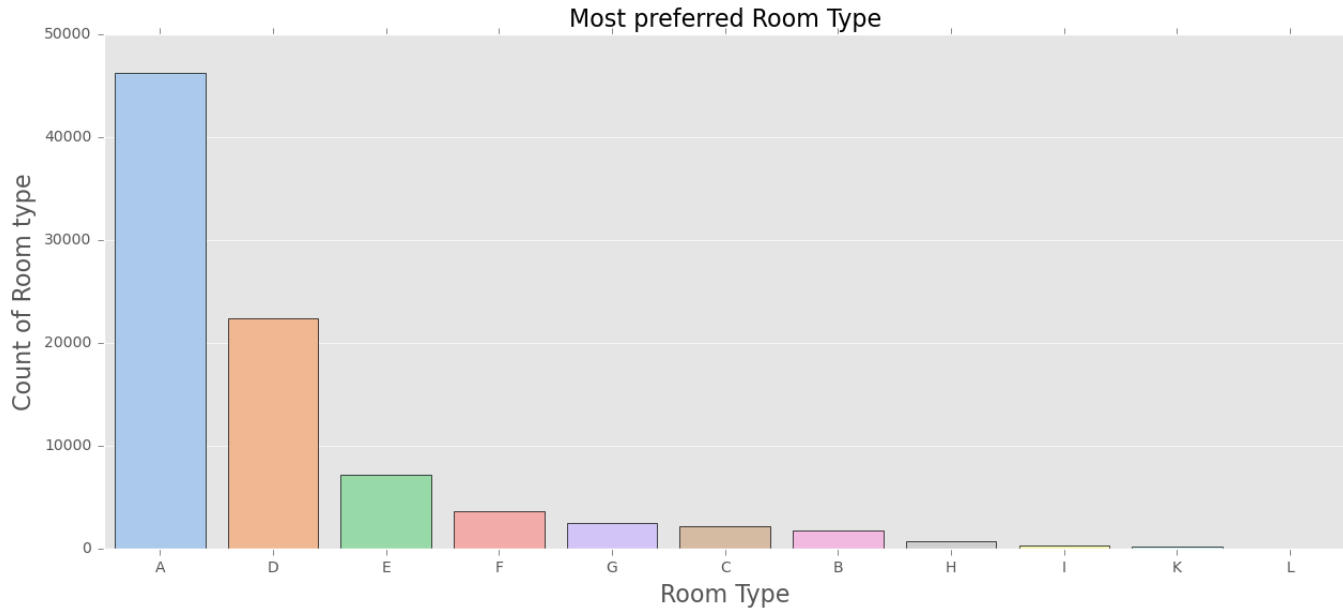


Insights found

From the chart, we got to know that **City Hotel** is **most preferred hotel** by the guests. Thus **City Hotel** has **maximum bookings**. **61.1% guests** are preferred **City Hotel**, while only **38.9% guests** have shown interest in **Resort Hotel**.

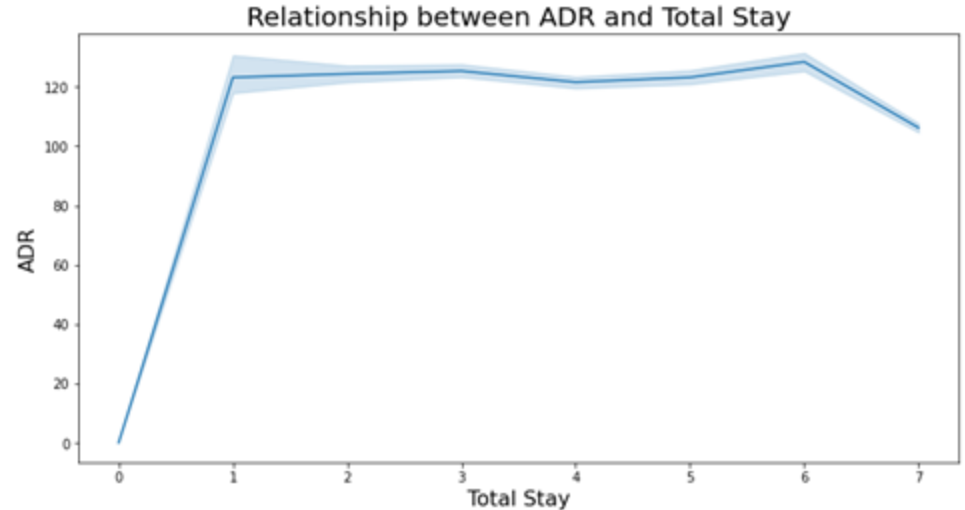
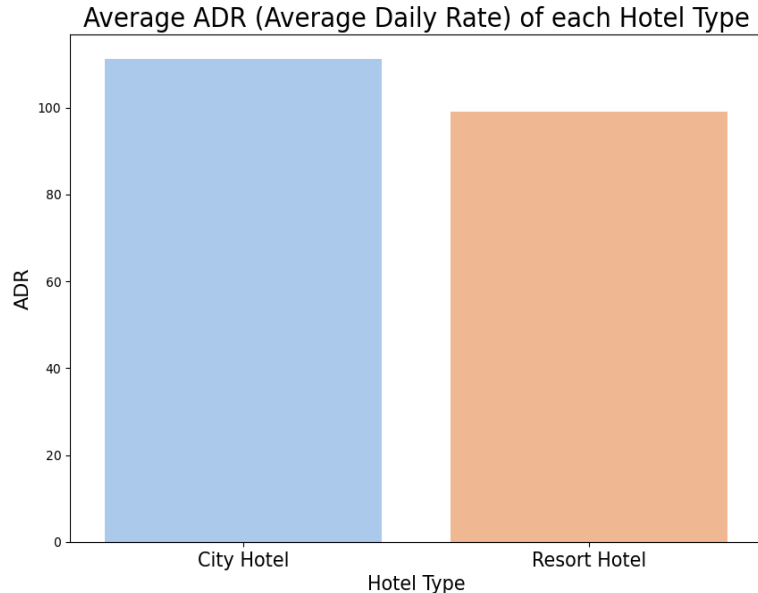
Most Preferred Hotel Rooms

- It is found that the **most preferred Room type** is 'A'. So, majority of the guests have **shown interest** in this **room type**.
- There are **positive impacts** because 'A', 'D', 'E' is **more preferred** by **guest** due to **better services** offered in room type.



Average Daily Rate

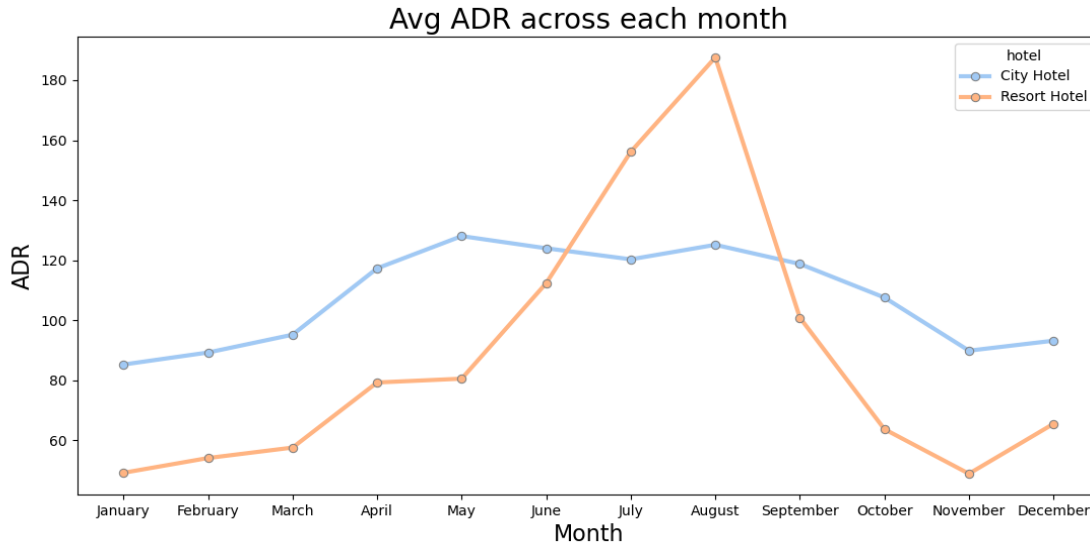
- **City Hotels** are generating **more revenues** than the **Resort Hotels**, because **City hotel** has the **highest ADR**. More the **ADR**, more will be the **revenue**.
- From the **line chart**, we have found that as the **total stay** increases the **ADR** is also getting **high**. So, **ADR** is directly **proportional** to **total stay**.



Average Daily Rate

➤ ADR across months

- For Resort Hotel, ADR is high in the months of June, July, August as compared to City Hotels. The reason may be that customers/people want to spend their summer vacation in Resort Hotels.

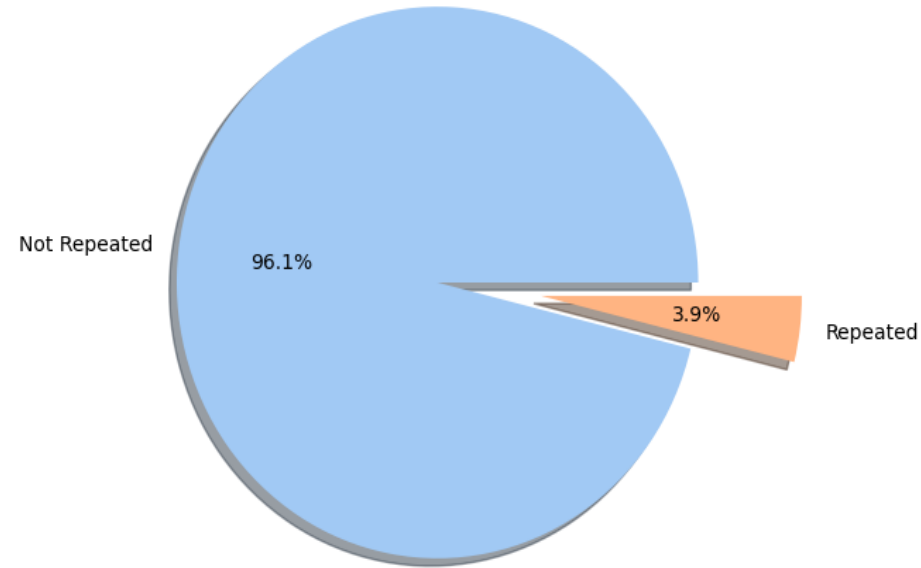


Repeated Vs Non-Repeated Guests

Percentage (%) Of Repeated Guest

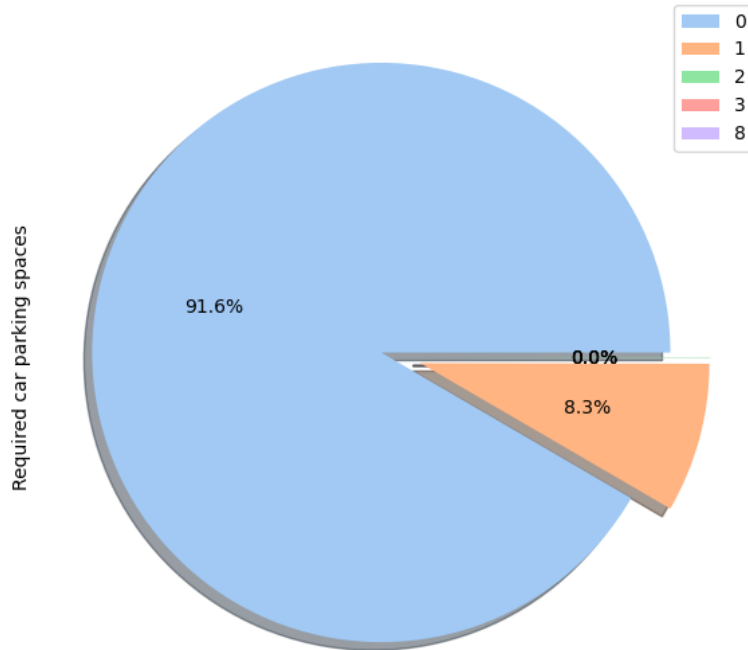
Insights found

- The pie chart show the **percentage** of **repeated guests** or **not** (where **0** is **not repeated guest** and **1** is **repeated guest**)
- **Repeated guests** are very few which is only **3.9%** while **96.1%** guests are **not returning** to the same hotel.
- The **guests management** should take **feedbacks** from guests and try to **improve** the **services**.



Requirement of Car Parking Space

% Distribution of
required car parking spaces



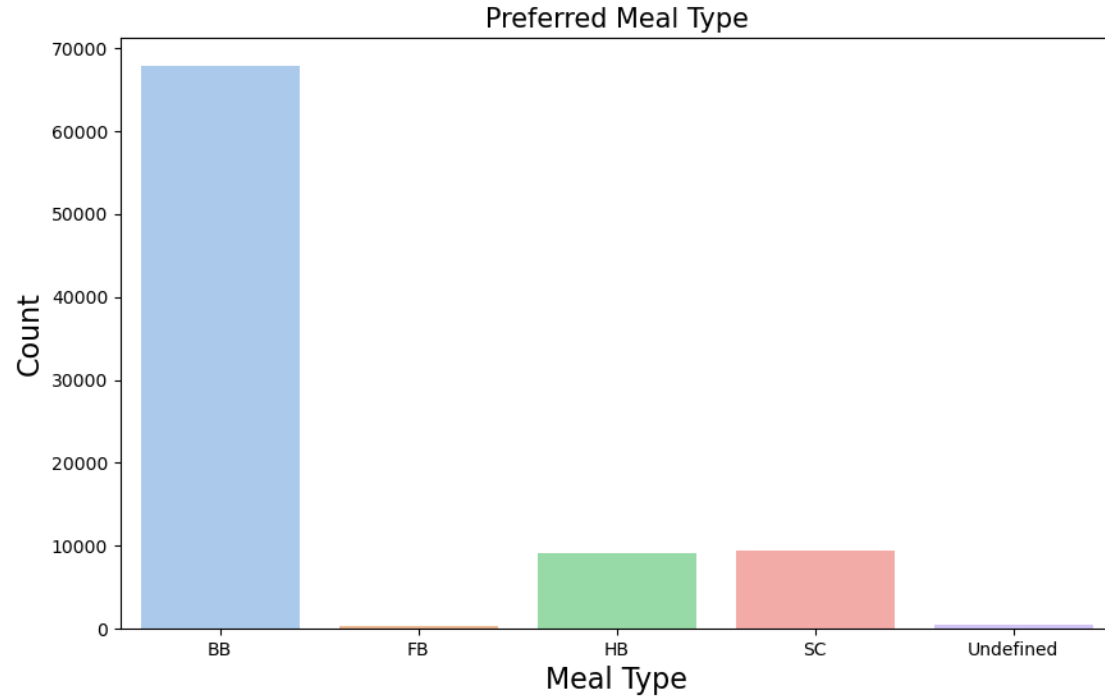
Insights found

- This chart shows that **91.6%** guests did **not required** the **parking space**. Only **8.3%** guests **required** the **parking space**.
- The **demand** for car parking area is **less**. It can be said that hotels need to **work less** on car parking spaces as only **1 car parking space** was required by **8.3%** of guests.

Most Preferred Meal

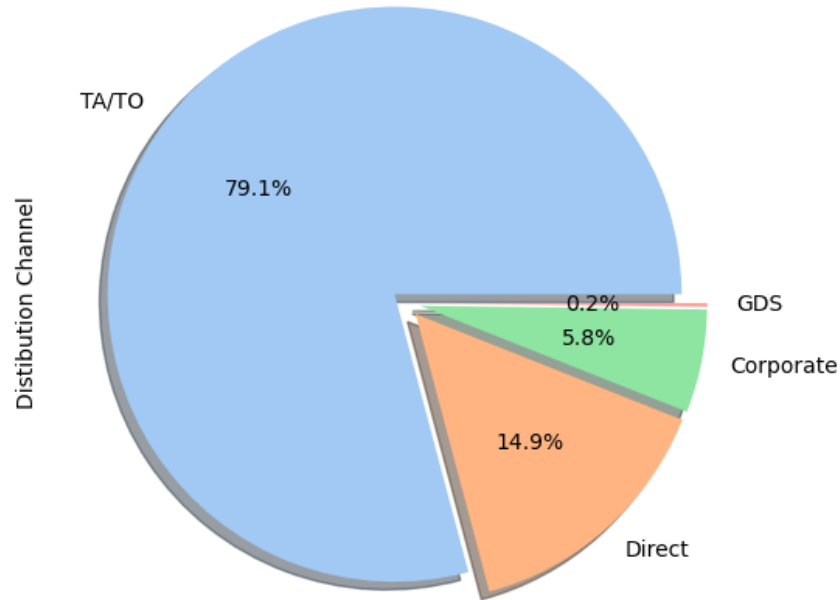
Insights found

The most preferred meal type by the guests is **BB (Bed and Breakfast)** while **HB (Half Board)** and **SC (Self Catering)** are equally preferred.



Maximum used Distribution Channel

Mostly used Distribution Channel
for Hotel Bookings

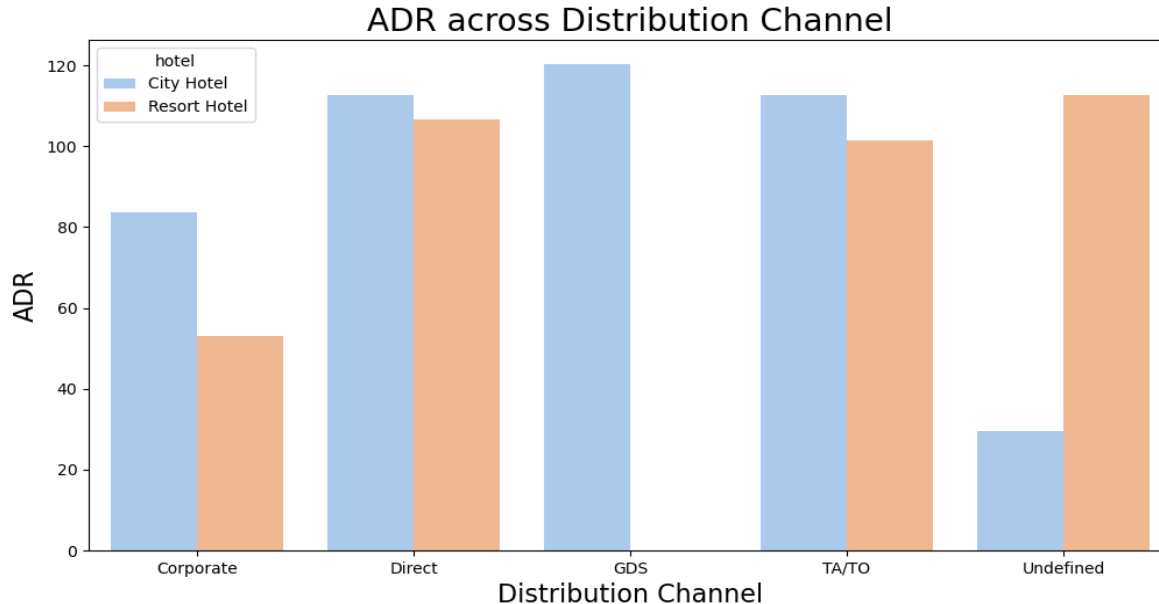


Insights found

'TA/TO' has been **mostly (79.1%)** used for **booking hotels**. **Direct** market segment of **14.9%**, **Corporates** market segment of **5.8%**, **GDS** market segment of only **0.2%** and rest **unidentified** are **0%**.

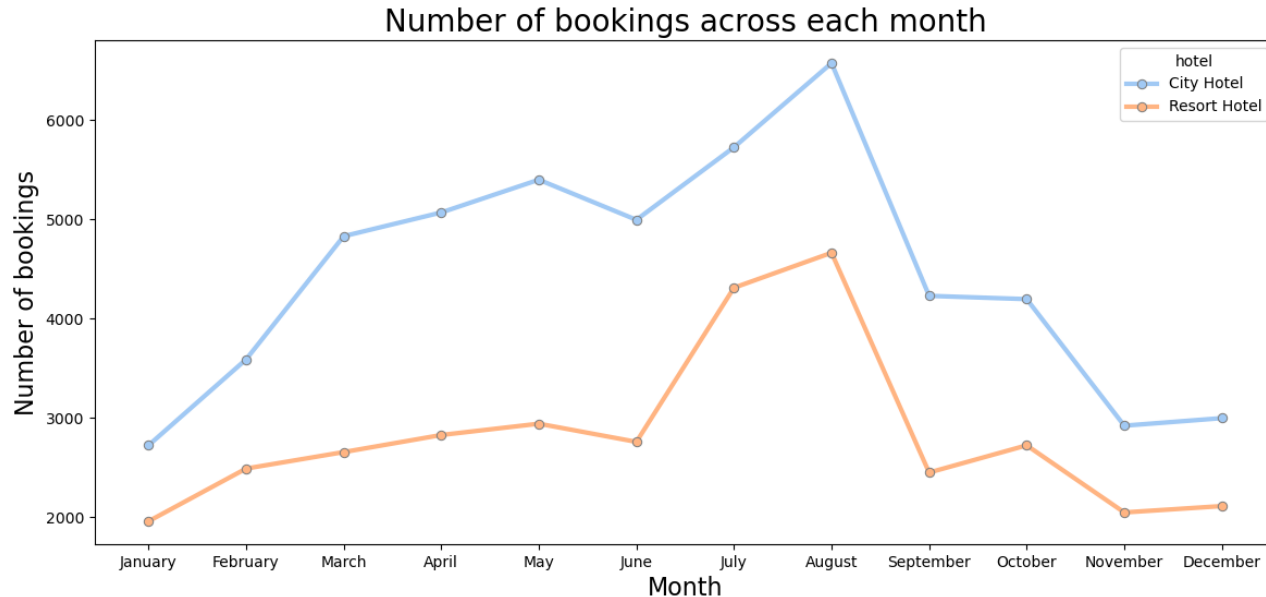
Distribution Channel Vs ADR

- **'Direct'** and **'TA/TO'** have almost **equally contribution** in **ADR** in both type of hotels. While, **GDS** has **highly contributed** in **ADR** in **'City Hotel'** type. **GDS** needs to **increase Resort Hotel** bookings.



Top Booking Months

- From this graph, we can say that **July** and **August** months had the **most Bookings**. As, **July** and **August** generally surrounds in and near the **summer vacation**.
- Hotels should be **well prepared** for the month of **July** and **August** as **maximum bookings** takes place for this **month**.



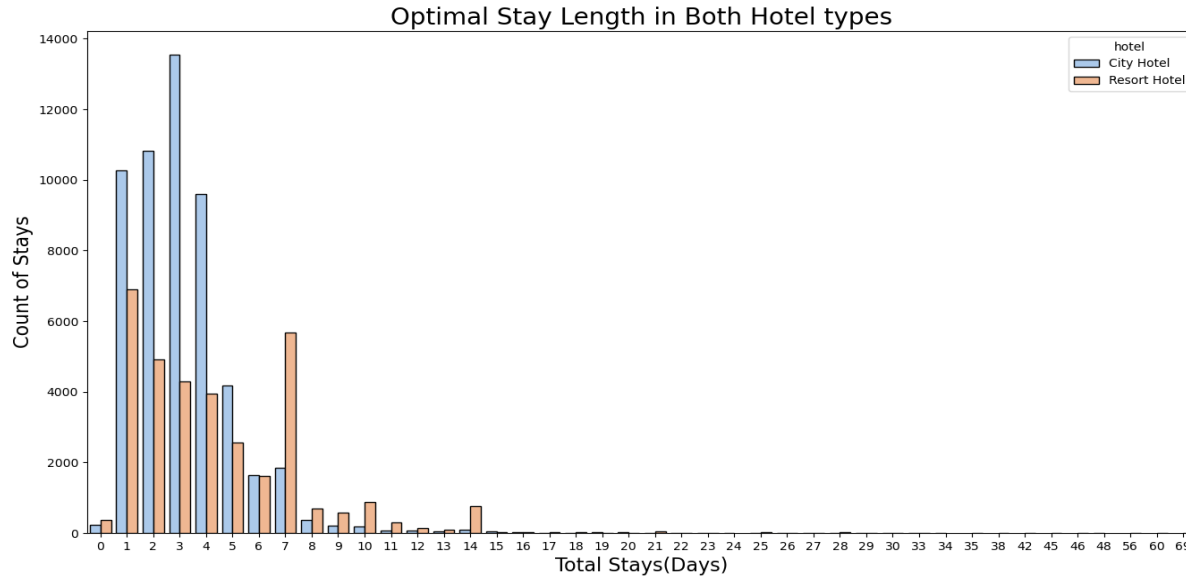
Top Booking Year

- It can be summarised that in the year **2016** both the hotel saw a **massive increase** in their **bookings** and by far the year **2016** is the year of the **highest bookings** of **both hotel**. In **2016** and **2017** the **City hotel** is having the **highest** number of **bookings** but in **2015** the **Resort hotel** is having the **highest** number of **bookings**.



Optimal Stay Length

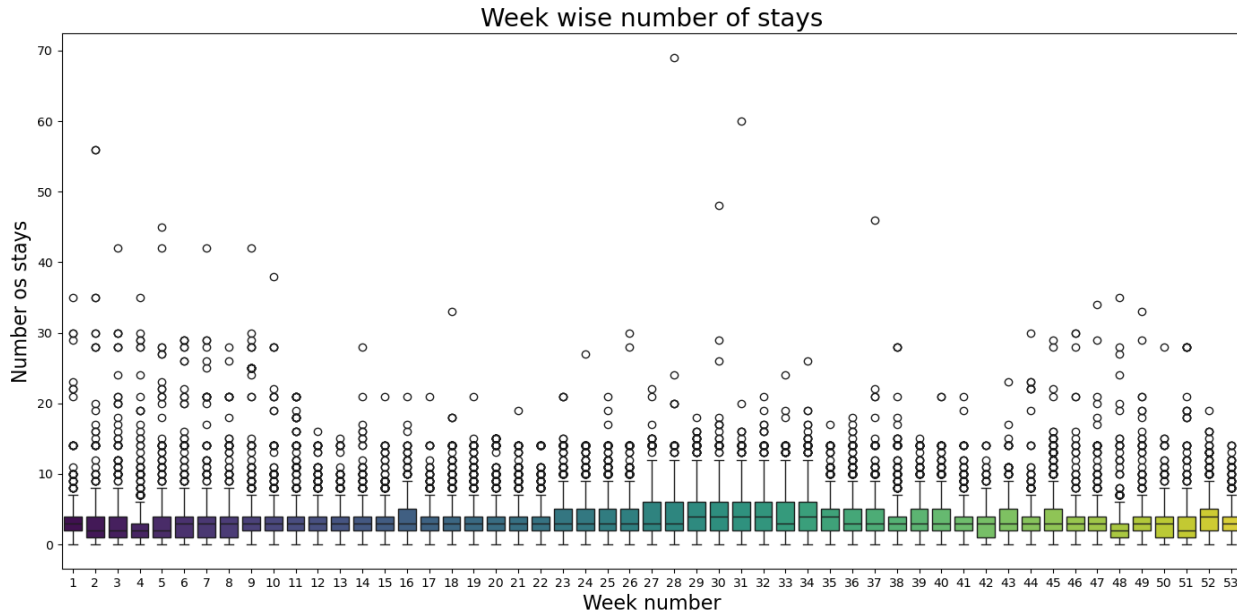
- We have found that the **Optimal stay** in both the type hotel is less than **7 days**. So, after that **staying numbers** have **declined** drastically.
- Customers usually **prefers** a **one week stay** in a hotel. So, hotels **need to work** efficiently in these **seven days** so that customers would **return** to the **same hotel** again.



Optimal Stay Length

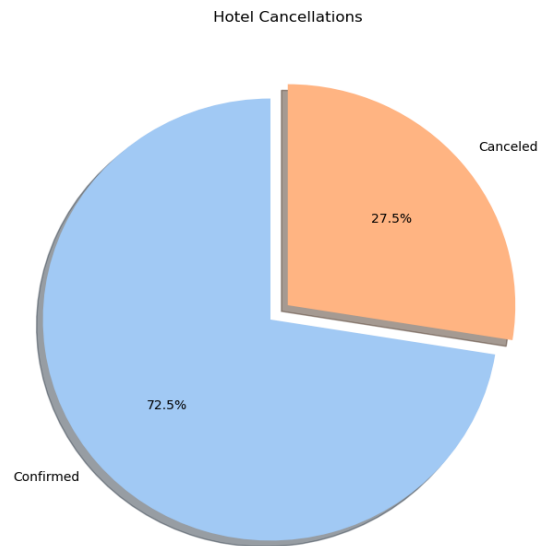
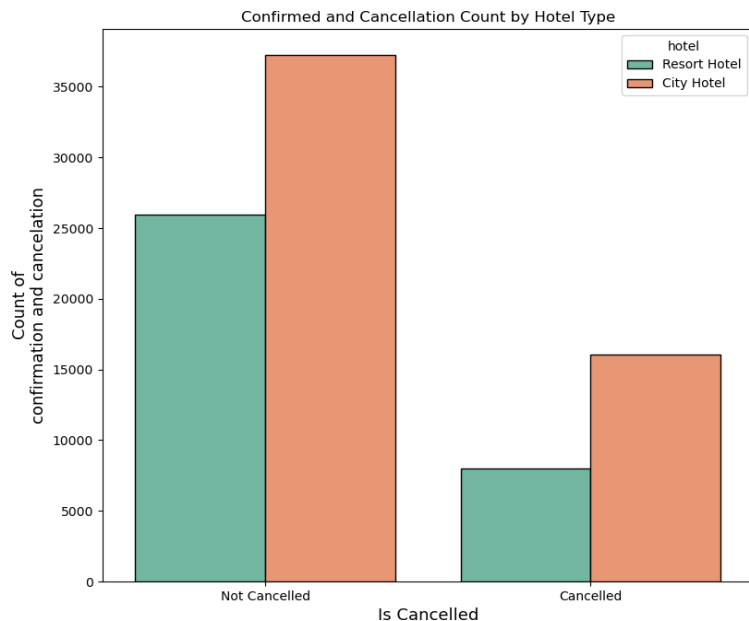
➤ Week Wise Number of Stay

- From the week **28 to 31**, it has shown the **highest days of stay** whereas from the week **1 to 11** has shown a very steady trend in the **number of stays** and also the week **18 to 22** has shown the **least number of stays** by the visitors in aggregate of all **3 years 2015, 2016 and 2017**.



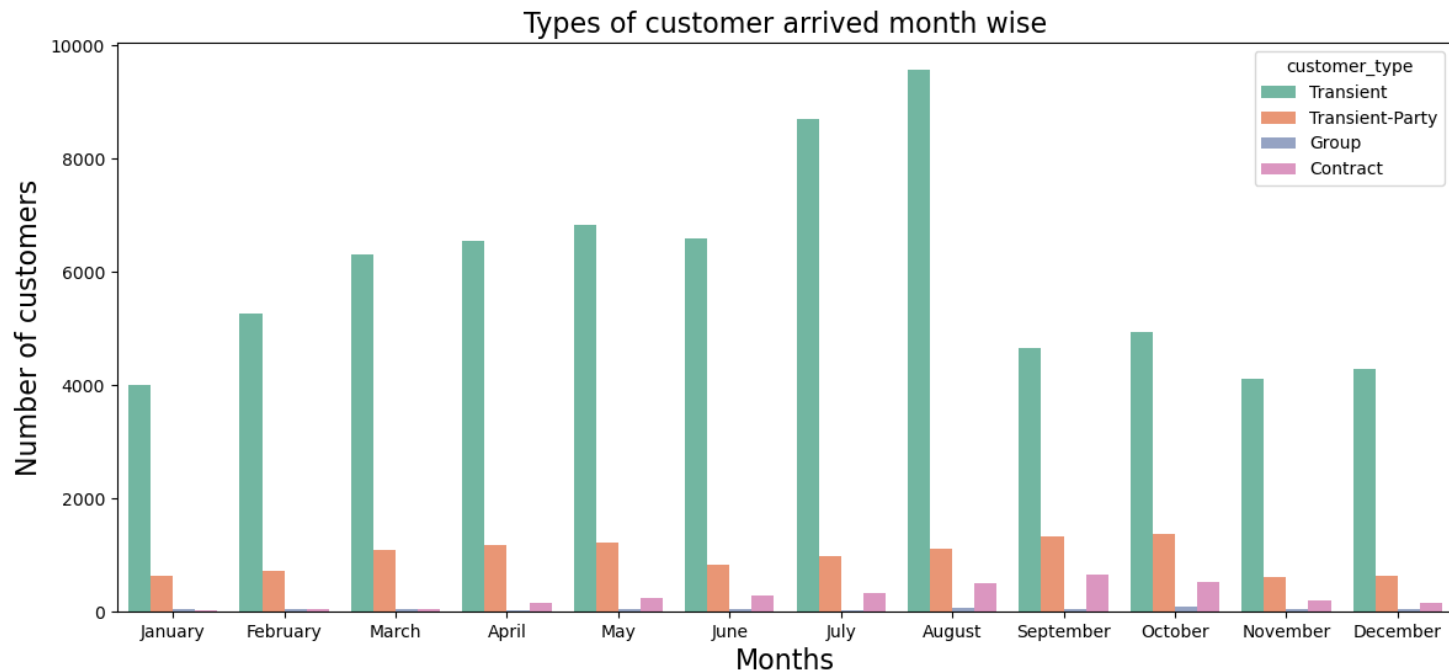
Confirmation Vs Cancellation

- More than **1/4th** of the **overall bookings** i.e. approx. **27.5%** of the tickets was got **canceled**.
- We can clearly deduce from the 2nd graph that the **City hotel** is having **greater** number of **bookings** as compared to **Resort hotel**. But, the **cancellation percentage** is also **high** of the **City Hotel**.



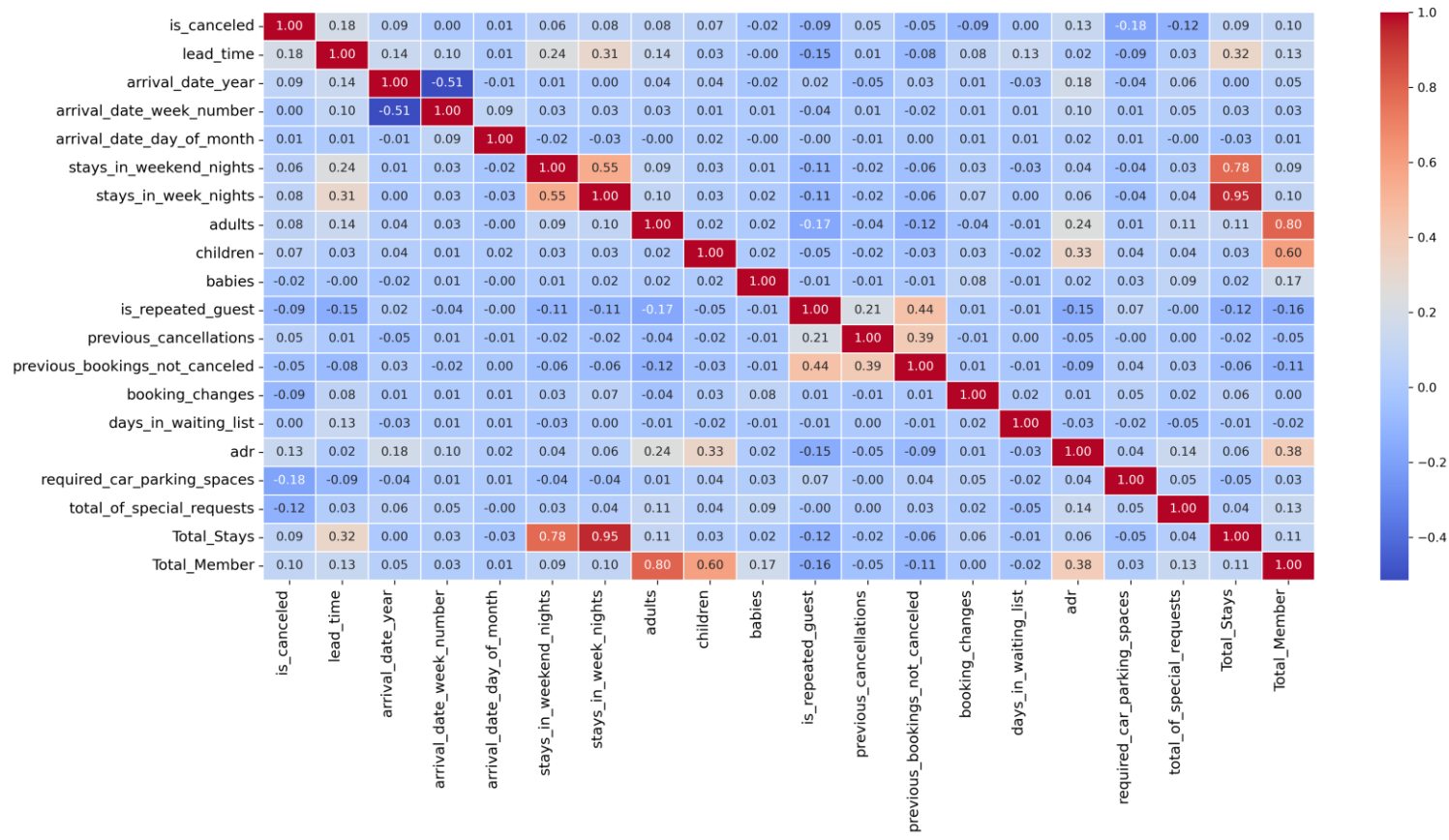
Mostly Arrived Customers/ Visitors

- It can be summarized that the **Transient** type of **customers** visit the **most** whereas the visitors who are in group comes in the category of **least visitors**.



Overall Stats (Co-relation Heatmap)

Correlation of Numeric Columns



Insight Found from the Heat Map

❖ Highly correlated:

- **'stays_in_weekend_nights' and 'stays_in_week_nights':** A strong positive correlation of 0.55 suggests that longer weekend stays are often associated with longer weeknight stays.
- **'total_of_special_requests' and 'Total_Stays':** A positive correlation of 0.78 indicates that guests making more special requests tend to stay longer.
- **'Total_Stays' and 'Total_Member':** A positive correlation of 0.95 suggests that the number of stays is highly related to the number of total members in the group.

❖ Moderately correlated:

- **'lead_time' and 'stays_in_week_nights':** A positive correlation of 0.31 indicates that bookings made further in advance are more likely to involve longer weeknight stays.
- **'lead_time' and 'Total_Stays':** A positive correlation of 0.32 suggests that longer lead times are associated with more stays overall.
- **'days_in_waiting_list' and 'Total_Stays':** A positive correlation of 0.38 implies that a longer waiting list might indicate a longer stay.

❖ Weakly correlated:

- **'is_canceled' and 'lead_time':** A positive correlation of 0.18, suggesting a slight tendency for cancellations to occur with longer lead times.
- **'adults' and 'Total_Stays':** A positive correlation of 0.11, indicating a slight tendency for more adults in a group to be associated with a longer stay.

Conclusion

- ❖ **City Hotels Are Most Preferred:**

City hotels are the busiest compared to resort hotels, making them the most preferred hotel type by guests.

- ❖ **Higher Revenue for City Hotels:**

The average ADR (Average Daily Rate) for city hotels is higher than that of resort hotels, indicating that city hotels generate more revenue.

- ❖ **Total Stay and ADR Relationship:**

The total stay of guests is directly proportional to ADR. Longer stays result in higher ADR and increased revenue.

- ❖ **Low Guest Retention:**

Only **3.9%** of guests are repeat customers, while **96.1%** are new guests, indicating a low retention rate.

- ❖ **Low Demand for Car Parking:**

Only **8.4%** of guests require car parking spaces. This shows that limited parking availability does not significantly impact the business.

- ❖ **Preferred Meal Type:**

Among meal types, **BB (Bed & Breakfast)** is the most preferred by guests, indicating their inclination towards simple and convenient meal plans

Conclusion

❖ Distribution Channel Contributions to ADR:

1. **Direct bookings** and **TA/TO** (Travel Agents/Tour Operators) have nearly equal contributions to ADR for both city and resort hotels.
2. **GDS (Global Distribution System)** significantly contributes to ADR in city hotels.

❖ Optimal Stay Length:

Guests typically stay for **less than 7 days** in both city and resort hotels. Stays longer than a week show a steep decline in frequency.

❖ Popular Months for Bookings:

The majority of bookings occur in **July and August**, making these the busiest months for hotels.

❖ Most Used Distribution Channel:

The most utilized distribution channel is **TA/TO**, accounting for **79.1%** of total bookings.

❖ Seasonal ADR Trends:

For resort hotels, the ADR peaks during **June, July, and August**, while city hotels exhibit relatively stable ADR throughout the year.

❖ High Cancellation Rate:

Approximately **27.5%** of all bookings are canceled, indicating a significant number of lost opportunities.

❖ Most Preferred Room Type:

Room type **A** is the most preferred by guests, showing the highest demand among all room types.

Thank You